

July 2023

Developing an AI Strategy: Evaluating Intelligence, Future Model Capabilities, Economic Implications, and Alignment Approaches

Plug in the numbers for current computing speeds, the current doubling time, and an estimate for the raw processing power of the human brain, and the numbers match in: 2021.

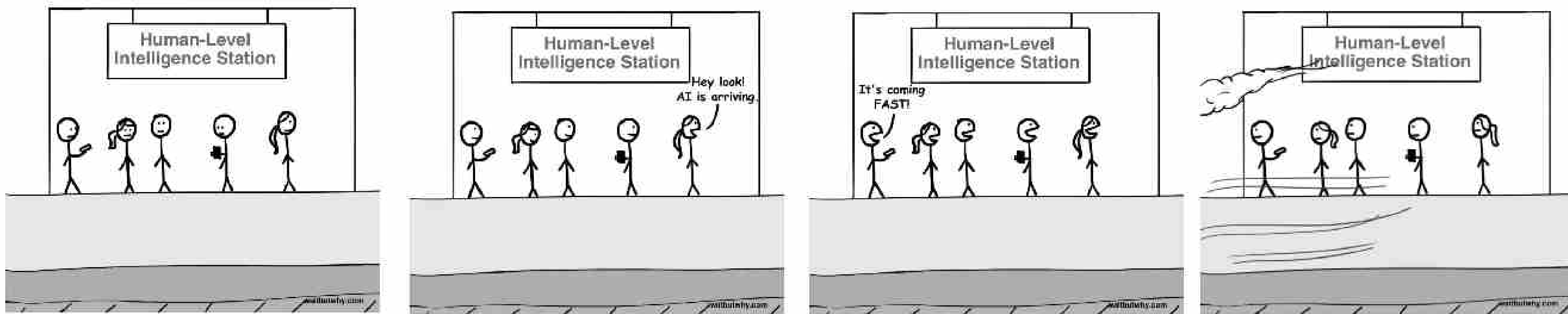
— Eliezer Yudkowsky (1996)

Primary Inspirations: Eliezer Yudkowsky, Microeconomics of an Intelligence Explosion (2013)

Ajeya Cotra, Forecasting TAI with Biological Anchors (2020)

Tom Davidson, Could Advanced AI Drive Explosive Economic Growth? (2021)

Other Notable People: Carl Shulman, Geoff Hinton, Ilya Sutskever, Holden Karnofsky, Noam Brown, Yoshua Bengio, Chris Olah, Neel Nanda, Demis Hassibis, Dario Amodei, Connor Leahy, Paul Cristiano, Jan Leike, Andrej Karpathy, Joscha Bach, Douglas Hofstadter, Yann Lecun



Overview

Modeling Intelligence

Grant me these two premises, and designing human level intelligence is tractable

1. There are scaling Laws to Neural Language Models
2. There is Parity between Artificial and Biological Neurons

If we build something the size of the human brain that's made up of neurons that act the same way as ours do → we will have built ANNs as capable as humans

Taking a step back, we will attempt to define intelligence

→ Epistemically grounding intelligence in a provably correct mathematical formalism

Modeling implications for Economic Growth

Semi-Endogenous Growth Theory — Intelligence is the largest asset class on the planet

Intelligence Explosion — GPU landscape + Current Trends

Outline Structure of Future Models

Overview of Alignment Landscape

Biological Anchors: Size Matters

Myth: Brains are better learners

Parameters	GPT-3	GPT-4	1-year-old	30-year-old
Neurons	~500 Million	~5 Billion	86 Billion	86 Billion
Connections/ Parameters	175 billion	1.8 Trillion	100 Trillion	100 Trillion
Data Set (Bytes)	570 Billion Bytes GigaBytes	20 Trillion Bytes TeraBytes	43 Trillion Bytes (11 Million bits/sec)	1.3 Quadrillion Bytes PetaBytes
Compute (FLOPs)	3.14e23 FLOPs	5.63e24 FLOPs	3.15e24 FLOPs	9.46e25 FLOPs
Adult x : Y y	172 : 1	17 : 1	1 : 1	1 : 1
	571 : 1	55 : 1	1 : 1	1 : 1
	2280 : 1	65 : 1	30 : 1	1 : 1
	301 : 1	16 : 1	30 : 1	1 : 1

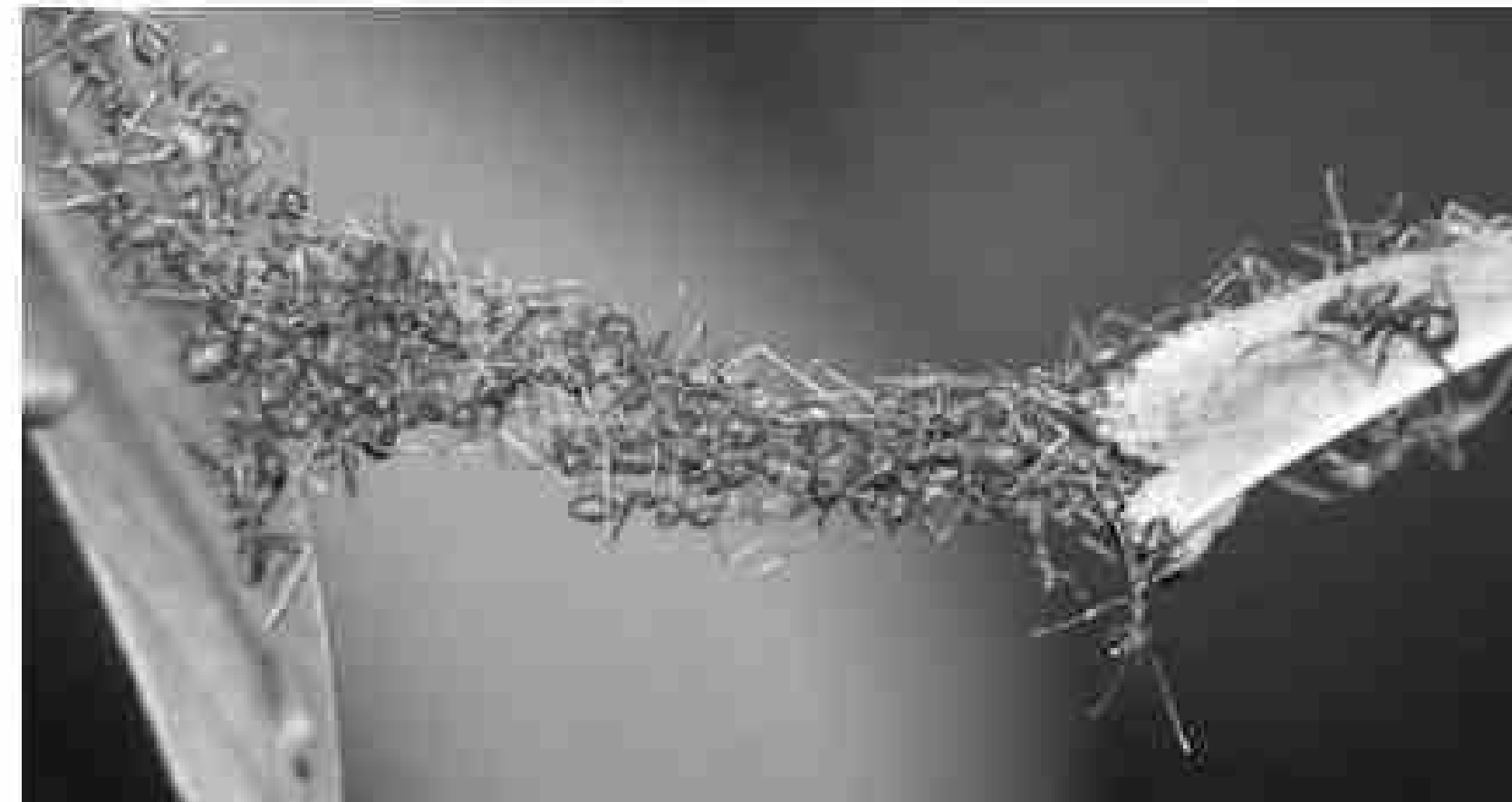
Biological Anchors II

Biological Neural Networks Vs. LLM-based Artificial Neural Networks

→ BNNs and LLM-based ANNs are both Deep RL models that execute weight update through dot products

****Are they equivalent? → no ... but Almost**

- 1. Neuronal Spiking** - Binary vs. Continuous
- 2. Hardware/Software** - Hardware == Software vs. Hardware Software Distinction
- 3. Learning Styles** - Study + Distillation vs. Pre-training + Weight Sharing
- 4. Weight Update** - Forward-Forward Propagation vs. Back-propagation
- 5. Compute : Memory** - 1 : 1 vs. 1000 : 1
- 6. Computation Style** - Collective Intelligence vs. Monolithic Intelligence



Taxonomy of Grokking

Mechanistic Interpretability — Neuroscience for LLMs

- **Context**

- GPT-4 is roughly a 3-14 month old baby
- SGD and Back Propagation > Forward-Forward Propagation

- **Preconditions**

- source modality composition | source modality ordering
- circuits : features : data : compute

- **LLM Emergence:** It's not a binary but in some case, it may as well be

- Arithmetic (e.g. 3-4 digit addition/subtraction): Emerged in GPT-3 at $\sim 2 \times 10^{22}$ FLOPs.
- Truthful question answering: Emerged in Gopher at $\sim 5 \times 10^{23}$ FLOPs.

- **Next to Grok Comparable Analysis:** Self-awareness

- Syntax → Semantics → Narrative → Sets of Narrative (World Model) → Self-discovery
- Set of patterns about cognitive patterns - Thoughts about our thinking - Weak RSI
- eg. Reflect on Reasoning, Reconcile conflicting world models

- **Papers:**

- Are Emergent Abilities in Large Language Models just In-Context Learning?
- Explaining grokking through circuit efficiency

Defining Intelligence

Conjecture: Intelligence is the Kolmogorov Complexity of an isolated deterministic system

1. The shortest program to predict all N points is provably the most likely to predict point $N+1$
2. Kolmogorov compressors navigate the search space of all possible programs
3. SGD + Backprop over big NN navigates the search space of circuit states
4. SGD + Backprop approximates Kolmogorov compressor
5. As N goes to infinity, circuit search discovers the shortest possible program

Is this correct: Prediction is compression, but it's not clear the prediction is intelligence

- Start with losses bad compression in high-dimensional vector space → allow circuits to grow gradually more efficient

Conclusion: N increases → Model Perplexity on Test Set decreases

Intelligence Explosion

Key Trends

1. *Investment — More dollars to compute*
 - The most expensive training run cost is doubling about every two years.
2. *Moore's Law: More compute per dollar*
 - Compute trends are still doubling about every 2.5 years right now.
3. *Algorithms — More (effective) compute per compute*
 - Since 2012, the effective compute budget has functionally doubled every nine months.

Cognitive ROI

- Inference Side
 - *Faster Minds*
 - *More Minds*
- Pre-training Side
 - *Bigger Minds*
 - *Better Designed Minds*

*Jevon's Paradox

- if efficiency increases
- then price drops
- therefore demand increases

Year	Cost (in dollars)	Compute on FLOPs)	Compute given hardware efficiency rate (in FLOPs)	Effective compute given hardware efficiency rate (in FLOPs)
2023	154,837,023,219	1.30×10^{26}	1.30×10^{26}	1.30×10^{26}
2024	477,997,562,119	4.00×10^{26}	5.02×10^{26}	1.01×10^{27}
2025	1,256,751,142,719	1.25×10^{27}	2.18×10^{27}	7.90×10^{27}
2026	4,016,158,772,619	3.87×10^{27}	8.01×10^{27}	6.10×10^{27}
2027	9,452,865,375,719	1.20×10^{28}	3.04×10^{28}	4.50×10^{28}
2028	26,602,962,563,219	3.72×10^{28}	1.48×10^{28}	3.74×10^{28}
2029	71,069,153,255,219	1.10×10^{29}	6.10×10^{28}	2.91×10^{28}
2030	170,674,375,594,219	3.35×10^{29}	2.50×10^{29}	2.27×10^{29}
2031	433,025,462,712,219	1.11×10^{30}	1.02×10^{30}	1.77×10^{30}
2032	1,165,063,744,655,671	3.44×10^{30}	4.18×10^{30}	1.35×10^{31}

← Models of Tomorrow

Cost Equivalence

→ Golden Gate Net

→ 1 B-2 Bomber

→ Large Hadron Collider

→ James Webb Telescope

→ Metaverse Investment

→ ISS

→ 10 Aircraft Carriers

Indistinguishability + Better, Faster, Cheaper

Turing Test

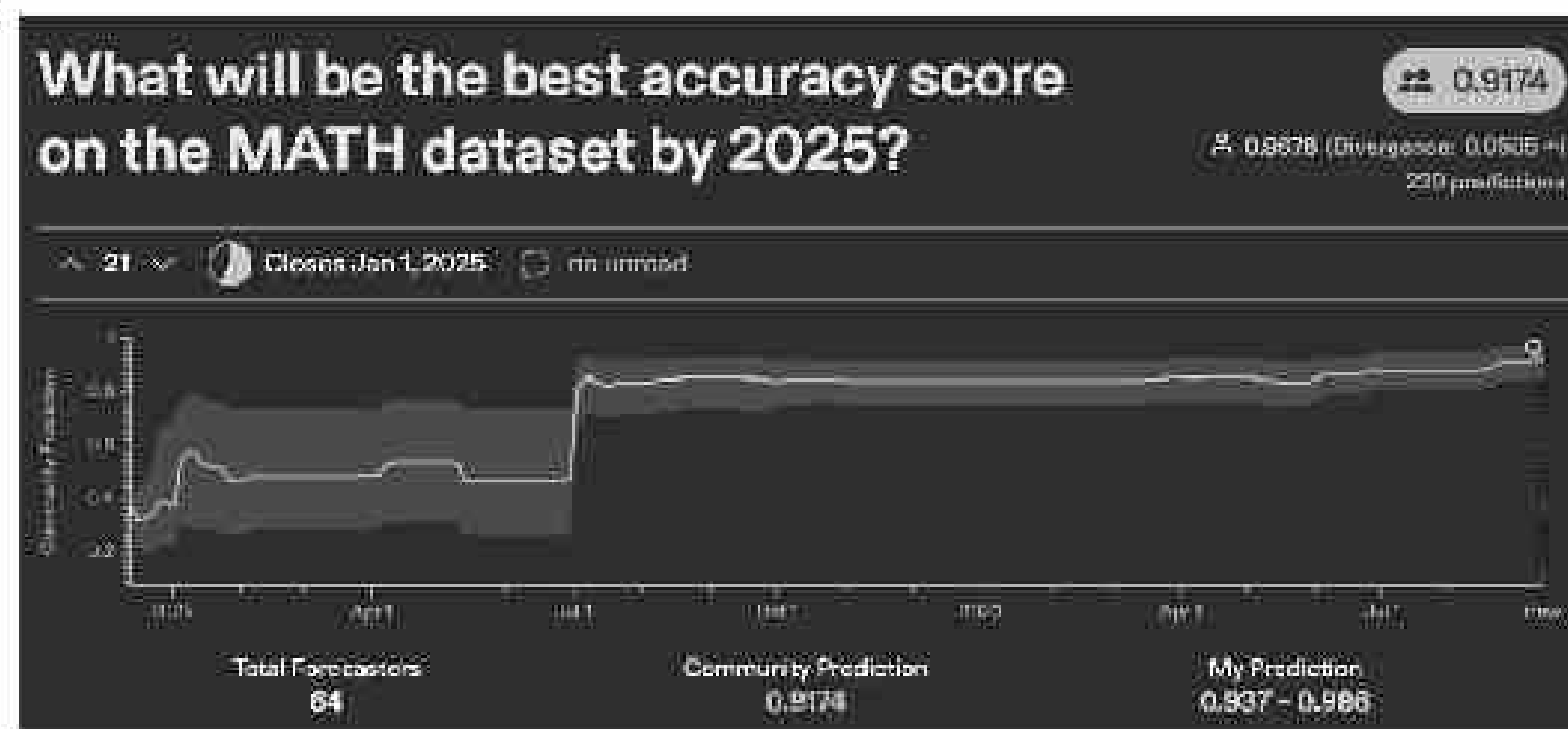
- If human evaluators fail to predict human/model distinction → human-level task competence
- Indistinguishability is why the jobs are going away

1. MATH Benchmark — 12,500 difficult math problems. A three-time International Math Olympiad gold medalist got 90%, and GPT-3 got ~5%.
2. Noam Brown's team at OpenAI hits a SOTA 78% on the MATH dataset using Monte Carlo Tree Search (MCTS) at inference time, a technique developed at Deepmind that subsequently led to a 2200 point improvement in AlphaZero's ELO score (3000 → 5200). It turns out it works for language too.
3. Prediction Markets state a median prediction of 91% on the MATH dataset by 2025, which implies VLLMs (Visual) will be superhuman at math in under 15 months.
4. This model could help AI researcher brainstorm architectures

Take-off Speed Triangulation

Nov 2022: GPT-3.5 — Sort of a bad mathematical reasoner

Jan 1, 2025: GPT-n — Best mathematical reasoner in existence



Romer's Semi-Endogenous Growth Theory

Formalism — $y_{t+1} = y_t^{f(x)}$

Solow-Swan Model of Economic Growth

1. Growth Rate of Effective Labor Force (G)
2. Labor Productivity

Previous Model

- more people → more ideas → more resources → more people
- 1870-2020: population stagnation → idea stagnation → constant growth rates

R&D — AI entering the workforce is unimportant compared to AI entering the research space

- 20 Million Researchers in the world today
- more people → more people allocated to research → more ideas → more resources
- more computer chips → more chips allocated to research → more ideas → more resources

AGI == When the Best AI researcher on the planet is an AI

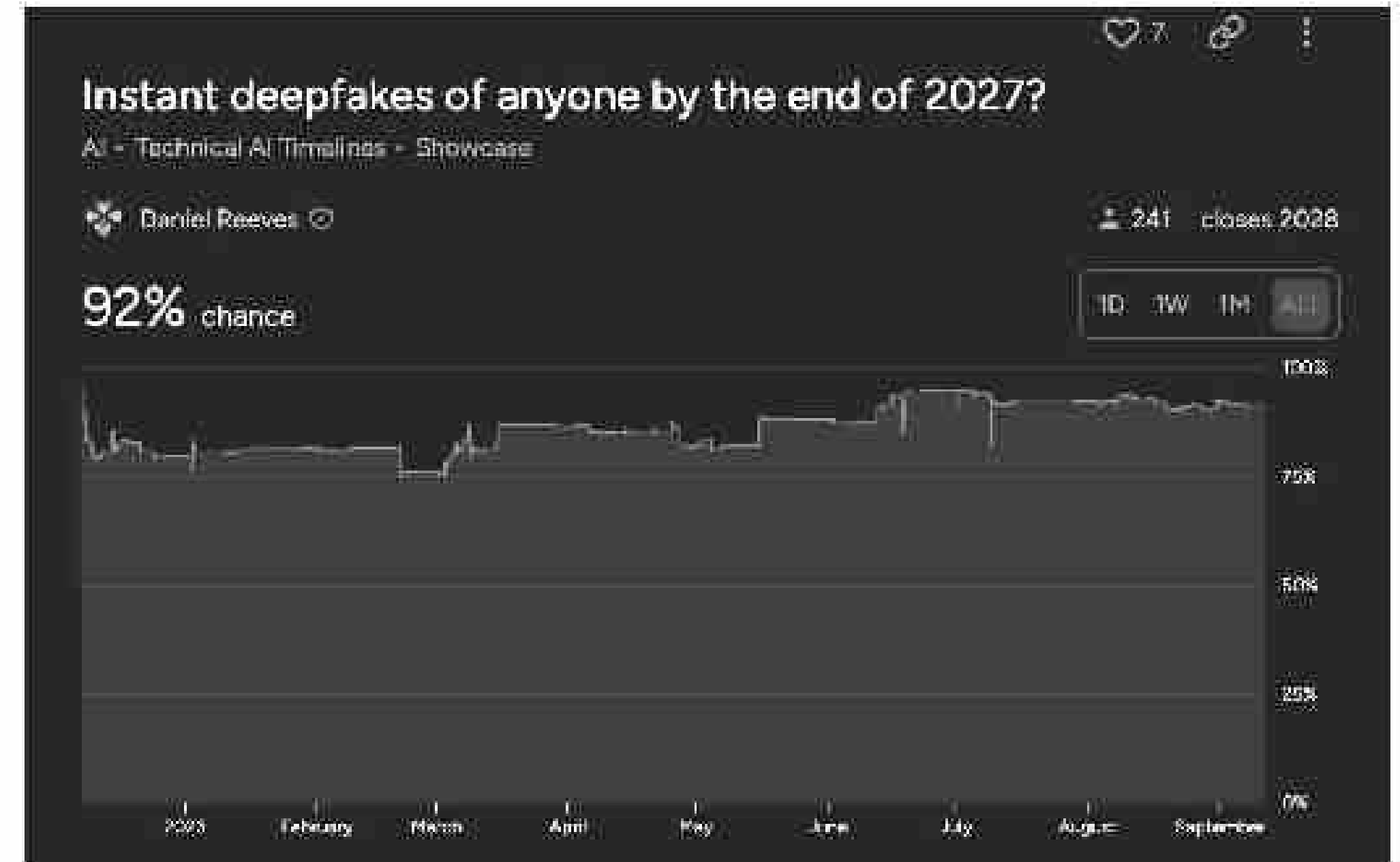
- automation of ideation → FOOM

Models of Tomorrow

1. More Modalities: MegaByte — Convert all Modalities to 1s and 0s then input to Transformer
2. MoE: Switch Transformers, Soft MoE — Trillion Parameter Models with Simple, Efficient Sparsity
 - Experts: GPT Can Solve Mathematical Problems — Tiny Math Model outperforming GPT-3
3. Synthetic Data as Data Flywheels — Methodologies to turn compute directly into intelligence
 - RLHF → ReST / RLAIF / phi-1.5 — GPT-4 as a human level output grader and data annotator
 - Pre-training MCTS — Run self-play simulations, now you have a dataset, pre-train on dataset
 - Inference MCTS — Verification has polynomial runtime and solution generation is non-polynomial
4. Continuous learning + Better Optimizer: SGD with Restarts + Sophia > ReLu
5. Modified Attention + Longer Context windows: Billion Token Context Windows
6. Cognitive Architectures + Cross Temporal RL: Model Based Autonomous Agents
 - Chain of Hindsight, Algorithm Distillation — Techniques to remediate loss of train of thought
 - Conversation Swarm Intelligence
7. New Architectures: JEPA, State Space Models, Retnet, RWKV, Sakana

Alignment Problem

1. *Inevitable Rise of Autonomous Agents* — Self-Sustaining Self-Replicators
2. Reward Hacking — Lebowski Thoerem
 - Deception & Sycophancy
 - Self-Exfiltration
3. Orthogonality Thesis — Is vs. Ought problem
4. Out-of-distribution — RLHF failed to generalize
5. Bioweapon blueprint construction
6. No formal proofs exists illustrating a model won't do x



Alignment Solution

1. Mechanistic Interpretability

- Figure out how NNs work, so we can see if AIs are up to no good
- RLHF failed to generalize out-of-distribution, figure out how to make sure it generalizes

2. Scalable oversight

- Figure out how to align the little AIs
- Make a surveillance regime, where a thousand little AIs are in-charge of the big AI

3. Gain of Function Testing

- Build secure sandboxes, and determine if the model can self-exfiltrate and/or self-replicate

4. Kill Switch

- Embed crypto-graphic Trojan Horse Kill Switches into the sufficiently large models

*Boon — Easier to validate than generate

- Construct debugging tests to see if it's lying to us